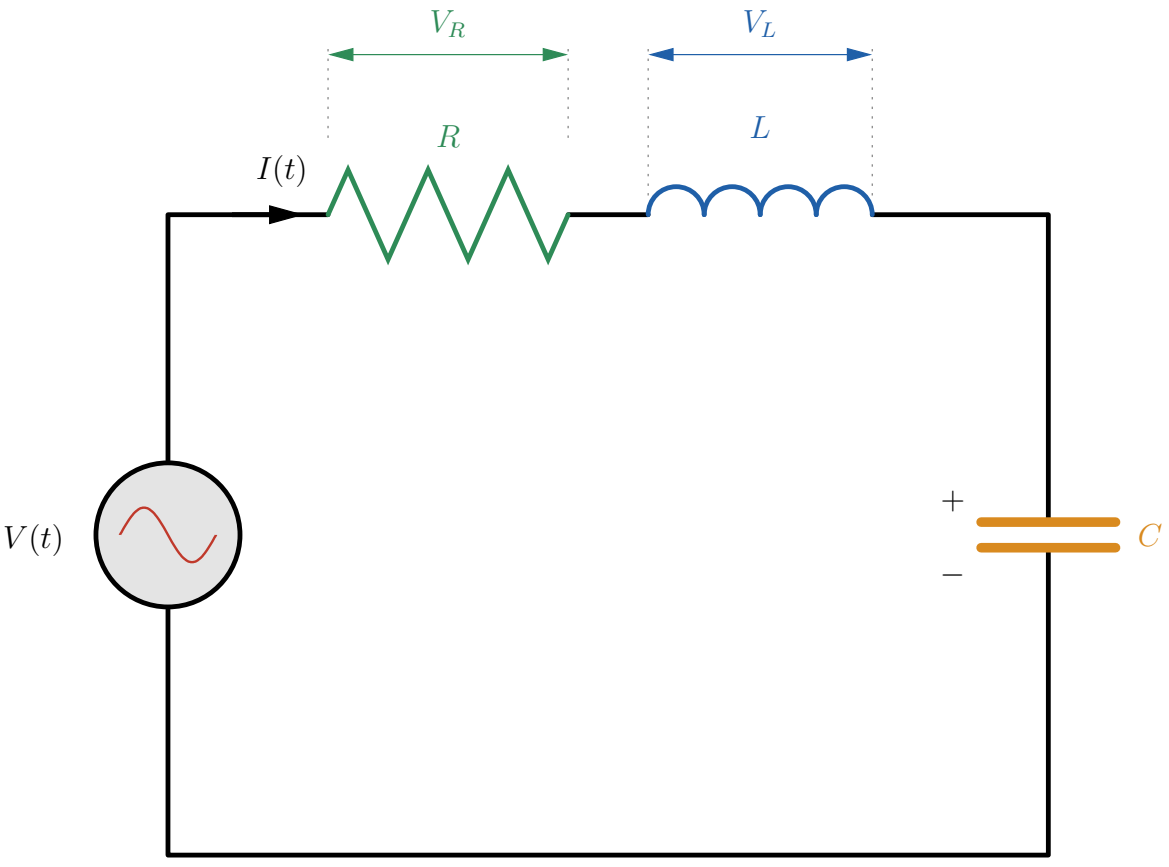
Series RLC circuit

AC source driving a resistor, an inductor and a capacitor in series



Kirchhoff's loop rule:

$$L\frac{dI}{dt} + RI + \frac{Q}{C} = V_0 \cos \omega t$$

$$\omega_0 = 1/\sqrt{LC}$$

resonance frequency

$$|Z| = \sqrt{R^2 + (\omega L - 1/\omega C)^2}$$